
Understanding Adolescent E-cigarette Use: Machine Learning to Explore Potential Risk Factors

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Abstract

The recent prevalence of adolescent e-cigarette usage has given rise to countless questions. Namely, many have tried to ascertain the factors that contribute to tobacco consumption. Previous research using machine learning to explore these factors has focused on demographic data, such as race, gender, income, etc. Nonetheless, this neglects a holistic perspective on the individual teen and their personal qualities, such as social media use, peers' perception, etc. Using data from the Center of Disease Control and Prevention's National Youth Tobacco Survey, this study (Github linked) uses models to determine whether or not the available data is sufficient for determining whether an adolescent is at risk for abuse based on relative amounts of tobacco consumption. The models include: logistic regression, decision trees, k -nearest neighbors (kNN), and support vector classifiers (SVC), with SVC achieving the highest accuracy of 53%. Our results demonstrate that there is not sufficient evidence to show that sparse holistic information and demographic data is able to predict risk levels of e-cigarette use amongst adolescents, which may help to inform how addiction prevention campaigns distribute their resources.

1 Introduction

The pervasive rise of e-cigarette use among adolescents in recent years has triggered widespread concern. As of 2022, about 1 in 10 students (middle and high school) in the U.S. have used e-cigarettes at least once in the past 30 days [6]. Vaping has adverse side effects for teens, including harming brain development, impacting cognitive abilities, and increasing susceptibility to other drug addictions [4]. In general, tobacco addiction is much more prevalent in some demographics than others. For instance, an article by Truth Initiative asserts that racism is a public health issue; the tobacco industry profiles African Americans, Hispanic and Latino communities, and other vulnerable populations, advertising in predominantly Black neighborhoods [10]. Lower-income communities are also less likely to be protected by smoke-free laws, making them more susceptible to secondhand smoke exposure [10].

Previous studies have confirmed that there is indeed a correlation between demographic data and e-cigarette use, with American Indian/Alaska Native students being most vulnerable to e-cigarette/cigarette use [7]. Further studies show that personal experiences are significant as well, with 79.1% of adolescent vapers having been introduced to e-cigarettes through their peers [1]. In this study, machine learning models will be used to assess the predictability of tobacco usage based on similarly selected factors. This study aims to determine whether or not demographic data and the sparse holistic information collected in national surveys are enough to make accurate classifications related to e-cigarette use. Using machine learning models, this research creates a classification problem; it labels whether adolescents are at risk for e-cigarette use or not. The analysis involves

Table 1: Correspondences between survey responses and predicted categories, including the frequencies of each category after preprocessing.

Days Used	Numerical Key	Label	Frequency
Never used + 1 day + 2 to 10 days (1.0)	0	Minimal usage	1,500
11 to 20 days + 21 to 50 days (2.0 or 3.0)	1	Mild usage	1,522
51 days to 100 days (4.0 or 5.0)	2	Moderate usage	984
Over 100 days (6.0)	3	Regular usage	637

implementing a collection of machine learning models to identify key predictors of adolescent e-cigarette use. The models used were Logistic Regression, Decision Trees, k -Nearest Neighbors, and Support Vector Classifiers (SVC). Given the correlations found in previous studies, we may expect that sparse and mostly demographic information should be enough for the models to learn trends in the data.

2 Dataset

In order to obtain data about adolescents’ tobacco usage across various factors, the main priorities are to (a) capture a diverse demographic, (b) use reliable sampling, and (c) be nationally-scalable. For this situation, it was determined that data from the Centers for Disease Control and Prevention’s (CDC) National Youth Tobacco Survey (NYTS) that is distributed to middle/high schools across the country would be most reliable. Since our study pertains to the recent surge in tobacco usage, we determined that the most recent 2022 survey would be best suited for our purposes.

To preprocess the data, there are three things to address: the type of questions/responses to train on, null values, and disproportionate data points. The columns selected to keep from the dataset (in order to provide a holistic view) corresponded to age, sex, specifications of race, healthcare visits, exposure to tobacco prevention campaigns, acknowledgement of nicotine/danger in tobacco products, peers’ perception of tobacco, exposure to tobacco advertisements, social media use, household tobacco usage, sexuality, gender, and frequency of tobacco use. Survey responses including null values were excluded from our training set. Nevertheless, there were a few exceptions in which the null values were replaced. For null values in the respondent’s age in years, we replaced these values with the mean across the dataset. For null values in sex, we replaced these values with a randomized response of male or female.

Next, to classify usage severity, we use the correspondences presented in Table 1. Numerical values were assigned to correspond to a risk classification, with 0 to 3 being from least to most addicted. It is important to note that the distribution of severity classification was imbalanced, with significantly more data points that indicated never having used e-cigarettes. To ensure that this distribution would not bias the models, data points were randomly removed from the “Minimal Usage” classification, leaving at least several hundred data points in each classification.

3 Methodology

As models will be trained on a labeled dataset, this study will utilize supervised learning. The dataset was trained on the specified X and y values: y represented the usage severity classification, and X represented the corresponding data in other columns. The training data comprised 80% of the provided data, and the remaining 20% was reserved for testing.

Logistic regression is a statistical and machine learning algorithm used for binary classification tasks, modeling probability of a discrete outcome based on predictor variables using a logistic function to produce results bounded between 0 and 1. The output of logistic regression is a probability score between 0 and 1, representing the likelihood of being at risk of use. Since logistic regression has relatively few hyperparameters, no further tuning was conducted.

kNN is used for classification and regression tasks; it predicts the label of a data point based on the majority label of its k -nearest neighbors in the feature space. The hyperparameter used in tuning kNN is the number of neighbors to compare to; from 1 to 25, the kNN’s accuracy, precision scores,

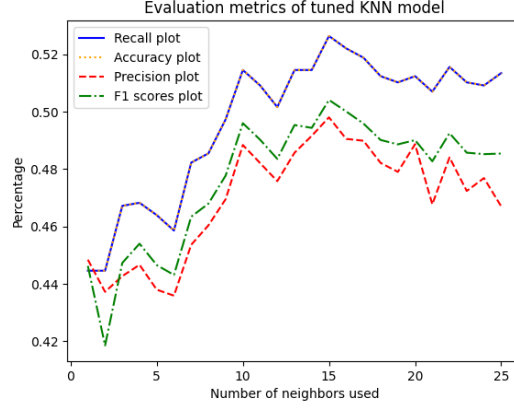


Figure 1: Evaluation metrics for trained kNN model. Note that the values of the evaluation metrics consistently peak at 15 neighbors.

recall scores, and F1 scores were recorded. Then, the evaluation metrics of the kNN were plotted to determine the optimal number of neighbors. For demonstrative purposes, the graph is included in Figure 1. From the graph, the model was most effective with 15 neighbors. Moreover, the function GridSearchCV, imported from sklearn.model_selection, used a parameter grid with 3, 5, 7, and 9 neighbors, different methods for calculating weights (either uniform or based on distance), different algorithms for finding neighbors (automatic selection, ball tree, k-d tree, and brute force), and different leaf sizes (10, 20, 30, and 40). Once the model was trained, it was determined that the best hyperparameters are: the k-d tree algorithm, a leaf size of 20, 9 neighbors, and uniform weights.

Decision trees are a machine learning algorithm used for classification and regression; model decisions and their possible consequences as a tree-like structure, with nodes representing decision points, branches representing outcomes, and leaves representing final classifications or values. To tune decision trees, we recorded the evaluation metrics from depths of 1 to 25, with the best max depth determined to be 5. GridSearchCV used a parameter grid of: "gini" and "entropy" for criterion to measure the quality of a split, "best" and "random" to choose the split, none, 5, 10, 15, and 20 for maximum depth, 2, 5, and 10 for minimum number of samples required at a leaf node, 1, 2, and 4 for the number of features to consider when looking for the best split, and "auto", "sqrt", "log2", and none for the number of features to consider when looking for the best split. It was determined that the best hyperparameters are: using "gini" to measure the quality of a split, a maximum depth of 5, no limit on the number of features, 2 samples at a leaf node, 10 samples required to split a node, and choosing splits randomly.

Support Vector Classification (SVC) is an algorithm used for classification tasks, operating by finding the optimal hyperplane that maximally separates different classes in the feature space. Recording evaluation metrics of c-values (or the regularization parameter) of 0.001, 0.01, 0.1, 1, 10, and 100, the best c-value was determined to be 1.

4 Results

Below are the confusion matrices for each untuned model. Note that the model is able to learn general trends for categories 0 and 3, but unable to cope with mid-range usage levels. Instead it has incredible difficulty in accurately predicting those in category 1 and is resistant to categorizing anybody into category 2. This suggests that we might find higher accuracy if we used binary classification (minimal vs high use), but such sweeping classifications would not be useful for practical purposes such as determining where to allocate resources for preventative campaigns.

As seen in Table 2, tuning manually using for loops (as discussed in Section 3), the model that performed most optimally was SVC (with C equal to 0.1). Our tuned SVC model had an accuracy of 55%, precision of 48.59%, recall of 54.57%, and F1 of 54.57%. The remaining models performed comparably; however, precision tended to dip below 50%.

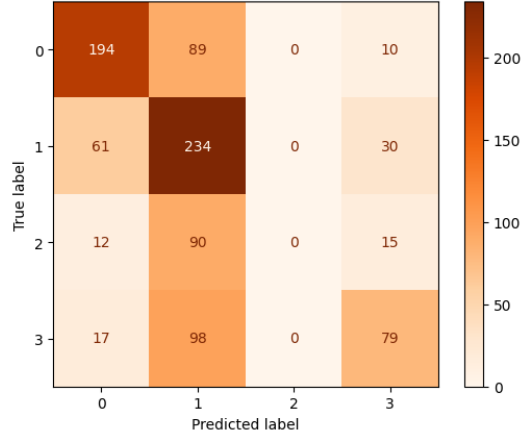


Figure 2: Confusion matrix for best performing model, kNN. Note that the model is especially strong at identifying label 0, and fails to identify those in category 2.

Table 2: Evaluation metrics of tuned models

Evaluation Metrics	Logistic Regression	kNN	Decision Trees	SVC
Accuracy	52.31	52.53	51	55
Precision	46.5	49.64	46	48.59
Recall	52.31	52.53	51.45	54.57
F1	48.91	50.32	48.15	54.57

Since sparse holistic data and demographic data was not able to generate strong predictions, there may be factors such as social trends outside of the particular subset of questions selected in this study. However, limitations of this approach include that the architectures used in this study are less complex. Consequently, relationships between the variables such as income and exposure to tobacco advertisements could have been missed. Moreover, this data, in spite of being anonymous, was still self-reported—adolescents may not feel comfortable admitting to usage. By removing null values, we may overlook insights from students who would’ve influenced the accuracy, but were not comfortable enough to answer a specific question. To address bias, training sizes of each label were considered. Since the labels of minimal/mild usage were significantly less than the labels of moderate/severe usage, data points in minimal/mild usage were omitted to match the number of data points in usage labels. However, this proved to have little impact on the accuracy of the model, with a difference of less than 1% accuracy. Although the low accuracy was unexpected, it only emphasizes the unpredictability of tobacco usage.

5 Conclusions

For this study, the research aim was to determine whether or not sparse personal qualities/experiences and demographic data is sufficient to determine risk levels for tobacco abuse. To do so, we analyzed four different machine learning models to determine if they would be able to effectively identify trends. With the highest accuracy of 53% (i.e. none outperforming random chance), we conclude that this data alone is insufficient to capture such trends. Thus, for effective preventative campaigns targeting the most vulnerable population, there will need to be a greater focus on robust holistic information. Future work may involve using more sophisticated models, such as neural networks, to determine if they’re able to pick up on trends. Ultimately, the results of this study’s conclusions may be used to help preventative campaigns, ensuring that they’re able to efficiently target their work.

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